Amenities Magnet: A tool to Identify Key Drivers of Residential Rental Prices in Germany (Team #146)

Terence B Buttrick (tbuttrick3), Ahmadou Diallo (adiallo41), Dionysios Siachras (dsiachras3),

Neerav Nemchand Gala (ngala8), Supraja Venkatesh (svenkatesh76)

Rental markets reflect complex interactions among property features, neighborhood conditions, yet listings and many predictive models present prices with limited transparency. Renters, investors, and policymakers therefore lack a clear explanation of what drives rent differences across locations. Amenities Magnet addresses this by predicting residential rents in Germany using ImmoScout24 listings while highlighting the most important features within each city to support clearer, locally grounded decision-making [5].

The objective is to develop an "Amenities Magnet" interactive dashboard that identifies and visualizes which apartment features most influence rental prices in a selected German city, using ImmoScout24 rental listings [7]. We use XGBoost (sequential boosted decision trees) to capture nonlinear relationships and interactions beyond linear regression, and use its learned split criteria to rank feature importance. The dashboard supports tenants, buyers, developers, and investors by clarifying price drivers and guiding design, renovation, and amenity investment decisions.

Current rental price prediction often relies on basic regression or black-box machine learning that outputs a price without showing which features matter [14]. Prior work suggests tree-based models can improve accuracy, including studies that combine standard listing details with listing text, though they mainly focus on prediction rather than explaining drivers [3]. Comparative research also finds ensemble methods perform well, but results are usually delivered as model outputs with little interactive exploration for end users [1]. Broader reviews confirm many modeling options, yet they are often region-specific and do not show how drivers change across cities [6]. Other studies note Random Forest and XGBoost capture nonlinear patterns better than older methods, but trust suffers when reasoning is not visible [9]. Related evidence documents a shift from hedonic models to data-driven ML, with tree ensembles often outperforming linear and kernel approaches, even though they are rarely turned into usable tools [10]. Since the best model is not always the most complex, we will benchmark multiple approaches across cities rather than assume one algorithm performs best everywhere [17]. Our project addresses these gaps by pairing predictive models with an interactive dashboard to explore feature importance across German cities.

Our approach provides deeper rental price prediction insights by offering a platform for region-specific modeling and feature selection, plus comparisons across multiple models. Prior work supports this: decision trees can capture key relationships with fewer inputs, while XGBoost often uses more variables and achieves higher accuracy, which matters for a method that is practical to run and still informative about what drives rent [2]. Explainable methods can break a prediction into factors that push rent up or down, but these are often shown as static figures rather than something users can explore, which our platform addresses [11]. Automated valuation research also finds ensemble and hybrid models outperform traditional hedonic baselines and increasingly pair accuracy with explanation tools such as SHAP, supporting our multi-model, explanation-first design [16]. Success is measured by (1) building several viable regional models and (2) delivering a platform that clearly communicates results for user decision-making.

Several groups benefit from understanding rental price drivers. City-level German studies show interpretable modeling can reveal clear, non-straight-line rent patterns, with size and building age often emerging as major drivers, supporting a way for tenants and buyers to see what shapes prices in their city [12]. Multi-city German research also shows these drivers change across locations as city composition and neighborhoods evolve, reinforcing the value of cross-city comparisons for decisions in different markets [13]. More broadly, spatial evidence finds the same feature can matter differently across geography, which helps planners and policymakers assess affordability and the influence of infrastructure and neighborhood conditions across cities [15]. Tenants can judge whether listings are fairly priced and

which amenities justify the cost, while investors and developers can target features with stronger returns for design, renovation, purchase, or upgrades. Planners and policymakers can use the results to understand how city and neighborhood factors relate to affordability across locations.

If widely used, this approach could improve rental price discovery and indirectly improve property valuation, since rental-income pricing models would be more accurate. Research on market shifts and shocks shows that housing prices can be shaped by forces like institutional investment and major events, and that patterns in neighborhoods can be practical signals for understanding how markets move over time [4], [8]. By giving more people a clearer view of what drives rent in each city, the tool can reduce information gaps between renters, owners, and investors. That added transparency can support fairer, more independent transactions between parties and contribute to more efficient housing markets overall.

The main risks are that models may not generalize equally well across all German cities, and gaps in the dataset could hurt results, since broad reviews suggest performance can vary by region and method [6]. The dataset may also have missing fields or uneven coverage, requiring careful cleaning and testing. On the explanation side, feature-attribution methods can clarify what drives predictions, but they are often presented statically and are difficult to turn into a simple, useful interface [11]. While automated valuation studies show accuracy can be paired with tools such as SHAP, the payoff depends on communicating results clearly to end users, not just analysts [16]. The payoff is a tool that predicts rent and shows users why a property costs what it does for tenants, investors, and developers.

As the Immoscout24 data [7] is openly available on Kaggle and the dashboarding tool Streamlit is free to use. Therefore, the cost of the project is zero.

The project is expected to take about four weeks. Week 1 focuses on data cleaning, exploratory analysis, and feature engineering. Week 2 covers model development and evaluation, with an emphasis on building models that perform reliably rather than overfitting. Week 3 extracts feature-importance metrics and builds the interactive dashboard, while also comparing several model options. Week 4 is for system integration, testing, usability evaluation, and preparing final deliverables (report, poster, and presentation).

The midterm checkpoint will confirm the data is ready and the model beats a simple baseline, showing early accuracy gains and an initial ranking of key features, consistent with prior evidence that ensemble methods can outperform simpler approaches for rent prediction [1], [10]. The final checkpoint will verify accurate predictions, stable feature rankings, and a working interactive dashboard, since explainability is most useful when it is clear and paired with strong performance [11], [16]. Progress will be tracked through weekly milestones, accuracy improvements, and dashboard usability testing, with success defined by technical reliability and practical clarity for end users.

0.1 PLAN OF ACTIVITIES

Phase	Owner	Timeline
Proposal & Lit Review	All	1/18–3/6
Data Pipeline	D. Siachras	3/1–3/15
Algorithm Dev.	A. Diallo	3/1–3/15
EDA & Tuning	N. Gala	3/1–3/29
Visualization & UI	S. Venkatesh	3/15–3/29
Integration & Test	T. Buttrick	3/22–4/19
Reports & Deliverables	All	3/8–4/24

All team members have contributed a similar amount of effort.

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A LITERATURE REVIEW CONTRIBUTION MAPPING

Table 1 summarizes the allocation of reviewed papers among team members. Citation numbers correspond to the numbered references in the bibliography.

Table 1: Mapping of Team Members to Reviewed Papers

Team Member	Reviewed Papers
Supraja Venkatesh	[3], [1], [6]
Ahmadou Diallo	[5], [10], [11], [15]
Terence B Buttrick	[8], [4], [16]
Dionysios Siachras	[9], [17], [14]
Neerav Nemchand Gala	[2], [12], [13]